ELISA CASTAGNARI

s2795874@ed.ac.uk | LinkedIn | github.com/Elis0 | +44 7884 988 232

ACADEMIC

UNIVERSITY OF EDINBURGH Edinburgh, UK
PhD in AI for Biomedical Innovation (Roche industrial partnership) Jan 2025 - Exp Dec 2028
- **Research Focus:** Improving top-1 retrieval accuracy for biomedical entity mapping by 16% (proposed method and augmented dataset currently undergoing human evaluation).

ALMA MATER STUDIORUM UNIVERSITY OF BOLOGNA Bologna, Italy
MSc in Artificial Intelligence Sep 2022 - Dec 2024
110/110 cum laude
Erasmus+ Exchange at Ecole normale supérieure (ENS) Paris, within the MPRI (Master Parisien de Recherche en Informatique)
Relevant Coursework: Deep Learning and Machine Learning, Natural Language Processing, Image Processing and Computer Vision, Multi-Agent Systems, Social Network Analysis, Combinatorial decision making and optimization, Cognition and neuroscience, Statistical and Mathematical Methods for Artificial Intelligence.

ALMA MATER STUDIORUM UNIVERSITY OF BOLOGNA Bologna, Italy
BSc in Computer Science Engineering 101/110 Sep 2019 - July 2022

PROFESSIONAL EXPERIENCE

Inria, Research Institute of France Paris, France
Research Intern Dec 2023 - July 2024
- Enhanced diagnostic precision over prior State-of-the-Art Deep Q-Network (DQN) reinforcement learning baselines by designing LLM-based pipelines for anaemia subtype classification
- Enhanced model interpretability and clinical alignment by integrating physician-defined decision trees into advanced prompt engineering workflows

Casy Laboratory, University of Bologna Bologna, Italy
Research Intern Jan 2022 - May 2022
- Increased temporal stability under signal noise compared to classical system identification by developing recurrent and convolutional neural networks for nonlinear dynamic systems
- Optimized model performance and interpretability across complex, real-world signals by conducting systematic ablation studies on candidate neural architectures

Unibo Motorsport (sponsored by Ducati) Bologna, Italy
Engineer: E-Powertrain Division Jan 2021 - May 2021
- Accelerated sensor telemetry analysis during live vehicle tests by engineering a real-time C++ data acquisition dashboard
- Reduced powertrain fault detection response time during track simulations by collaborating with a cross-functional team of 10+ mechanical and electrical engineers

University of Rome "La Sapienza" Rome, Italy
Research Intern Mar 2018 - May 2018
- Quantified physiological and behavioural indicators of body ownership by leading cognitive neuroscience experiments using immersive VR environments
- Advanced experimental methodologies across 2 subsequent lab projects by delivering insights directly adopted by Sapienza

SELECT RESEARCH PAPERS

1. [LLMs4MI 2024] Exploring Differential Analysis of Anaemia with Large Language Models. LLMs4MI 2024 @ FLLM 2024, IEEE, November 2024, Dubai, United Arab Emirates. View Paper
Castagnari, E., et al. (2024).

2. [CLEF 2025] Multilingual and Nested Biomedical Named Entity Normalization via Candidate Retrieval and Lightweight Large Language Model Disambiguation. View Paper
D. Lain, A., **Castagnari, E.**, et al. (2025).

3. [CLEF 2025] Understanding Gut-Brain Interplay in Scientific Literature: A Hybrid Approach from Classification to Generative LLM Reasoning. View Paper
Lee, C., **Castagnari, E.**, et al. (2025).
- Ranked 3rd at the BioASQ challenge using a hybrid classification and generative LLM reasoning pipeline.
- Increased F1-score by 25% and recall by 40% for Named Entity Recognition (NER) to locate and identify genes.

PRESENTATIONS & RESEARCH TRAINING

OxML Summer School: 10-minute oral presentation and poster presentation. 2026
HealTAC - "Improving Digital Healthcare Solutions with Data Interoperability and LLMs" 2025
MI4H - Poster presentation - "LOINC Laboratory Diversifier: A Benchmark Dataset for Interoperability" 2026
ECCB Conference, Geneva - Co-authored Workshop: From Multi-Omics to Gene–Disease Discovery: Knowledge Graphs and LLM-Augmented Analysis — ECCB 2026 T8 2026

SELECT PROJECTS

Anemia Diagnosis with LLMs
- Outperformed standard clinical baselines in diagnostic pathway discovery by prompt-engineering GPT-4 and LLaMA on synthetic medical cohorts
- Automated guideline-aligned diagnostic workflows by validating LLM decision pathways against expert hematology standards

Steam Search Engine
- Achieved sub-second query latency over 5M+ Steam user reviews by developing an inverted indexing retrieval system using BM25 hybrid scoring
- Processed multi-label sentiment classification at scale by deploying a cloud-native DistilRoBERTa NLP pipeline on Google Cloud Platform (GCP)

Quantitative Portfolio optimisation
- Outperformed the FTSE MIB benchmark over an 8-year backtest (2016–2024) by building a quantitative portfolio optimization engine using asset clustering and Sharpe ratio maximization
- Optimized monthly risk-adjusted returns by deploying ARIMA and ensemble forecasting models for automated asset rebalancing

Hypertension Clinical Pathways Comparison in Generation Scotland
- Enabled semantic queries on multi-thousand-patient clinical datasets by constructing an RDF/OWL knowledge graph linking NICE guidelines to Generation Scotland data
- Evaluated treatment adherence across patient cohorts by implementing an LLM-driven SPARQL generation pipeline for automated graph querying

SKILLS

- **Core Competencies:** Knowledge Graphs, Agentic AI, Large Language Models (LLMs), Knowledge Retrieval & QA Systems, Semantic Web Technologies (RDF/OWL, SPARQL).
- **Programming & Libraries:** Python (expert), PyTorch, LangGraph, C, C++, Java, R, SQL.
- **Tools & Platforms:** Git, Docker, Azure, Google Cloud Platform (GCP).

LANGUAGES

- Italian (Native), English (Fluent), Spanish (Conversational).

AWARDS & HONORS

- **Roche Industrial Partnership & UKRI PhD Funding:** Awarded for PhD research in AI for Biomedical Innovation, University of Edinburgh.
- **Academic Merit Internship:** Selected for a prestigious paid stage at the Scuola Normale Superiore di Pisa.